CORE PURE / FURTHER CALCULUS

Integration by Substitution

Questions by topic 15 questions · 107 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Instructions

Integration by substitution is assumed knowledge from A Level Mathematics. This pack revises it at Further Mathematics level with new functions that A-level Further Mathematics introduces.

Questions

Question 1	*6 marks*	2
Question 2	*8 marks*	3
Question 3	*11 marks*	4
Question 4	*6 marks*	5
Question 5	*10 marks*	6
Question 6	*6 marks*	7
Question 7	*6 marks*	8
Question 8	*6 marks*	9
Question 9	*8 marks*	10
Question 10	*6 marks*	11
Question 11	*6 marks*	12
Question 12	*7 marks*	13
Question 13	*6 marks*	14
Question 14	*8 marks*	15
Question 15	*7 marks*	16

1. Show that

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}}\,\mathrm{d}x = \frac{16}{15}(7-4\sqrt{2}) \qquad \textbf{[6]}$$

2. **(a)** Use the substitution $u = 1 + x^2$ to show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_p^q \frac{(u-1)^4}{2u}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = A + B\ln 5$$

where A and B are constants to be found. **[5]**

3. Using calculus, find the exact values of

(i)

$$\int_{-2}^{0} \frac{1}{x^2+4x+8}\,\mathrm{d}x \qquad \textbf{[5]}$$

(ii)

$$\int_{0}^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}}\,\mathrm{d}x \qquad \textbf{[6]}$$

4. Show that

$$\int \frac{\sqrt{x^2-16}}{x^2}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c \qquad \textbf{[6]}$$

5. Using calculus, find the exact values of

(i)

$$\int_1^2 \frac{1}{x^2+3x+2}\,\mathrm{d}x \qquad \textbf{[4]}$$

(ii)

$$\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x \qquad \textbf{[6]}$$

6. Show that

$$\int \frac{x^2}{\sqrt{x^2+16}}\,\mathrm{d}x = \frac{1}{2}x\sqrt{x^2+16} - 8\,\mathrm{arsinh}\left(\frac{x}{4}\right) + c \qquad \textbf{[6]}$$

7. Show that

$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}}\,\mathrm{d}x = \frac{32\sqrt{6} - 30\sqrt{5}}{5} \qquad \textbf{[6]}$$

8. Using the substitution $u = \frac{2}{3}\tanh x$, evaluate

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x \qquad \textbf{[6]}$$

9. **(a)** Use the substitution $u = \sqrt{2x+1}$ to show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = \int_p^q \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = A - B\ln 2$$

where A and B are constants to be found. **[5]**

10. Use an appropriate substitution to find the exact value of

$$\int_0^{\pi/6} \frac{\cos x \cos 2x}{(1+\sin x)^{3/2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\sqrt{6}$, where a and b are rational numbers to be determined. **[6]**

11. Use an appropriate substitution to find the exact value of

$$\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\pi$, where a and b are rational numbers. **[6]**

12. Find the exact value of

$$\int_0^{\frac{1}{2}\ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9\tanh^2 x}}\,\mathrm{d}x \qquad \textbf{[7]}$$

13. Find the exact value of

$$\int_2^4 \frac{\sqrt{x^2-4}}{x^2}\,\mathrm{d}x$$

Give your answer in simplest exact form. **[6]**

14. Show that

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2+\cos\theta}\,\mathrm{d}\theta = 15\ln\left(\frac{3}{2}\right) - 6 \qquad \textbf{[8]}$$

15. Show that

$$\int_0^{\pi/2} \frac{\cos^5\theta}{2+\sin\theta}\,\mathrm{d}\theta = 9\ln\left(\frac{3}{2}\right) - \frac{41}{12} \qquad \textbf{[7]}$$